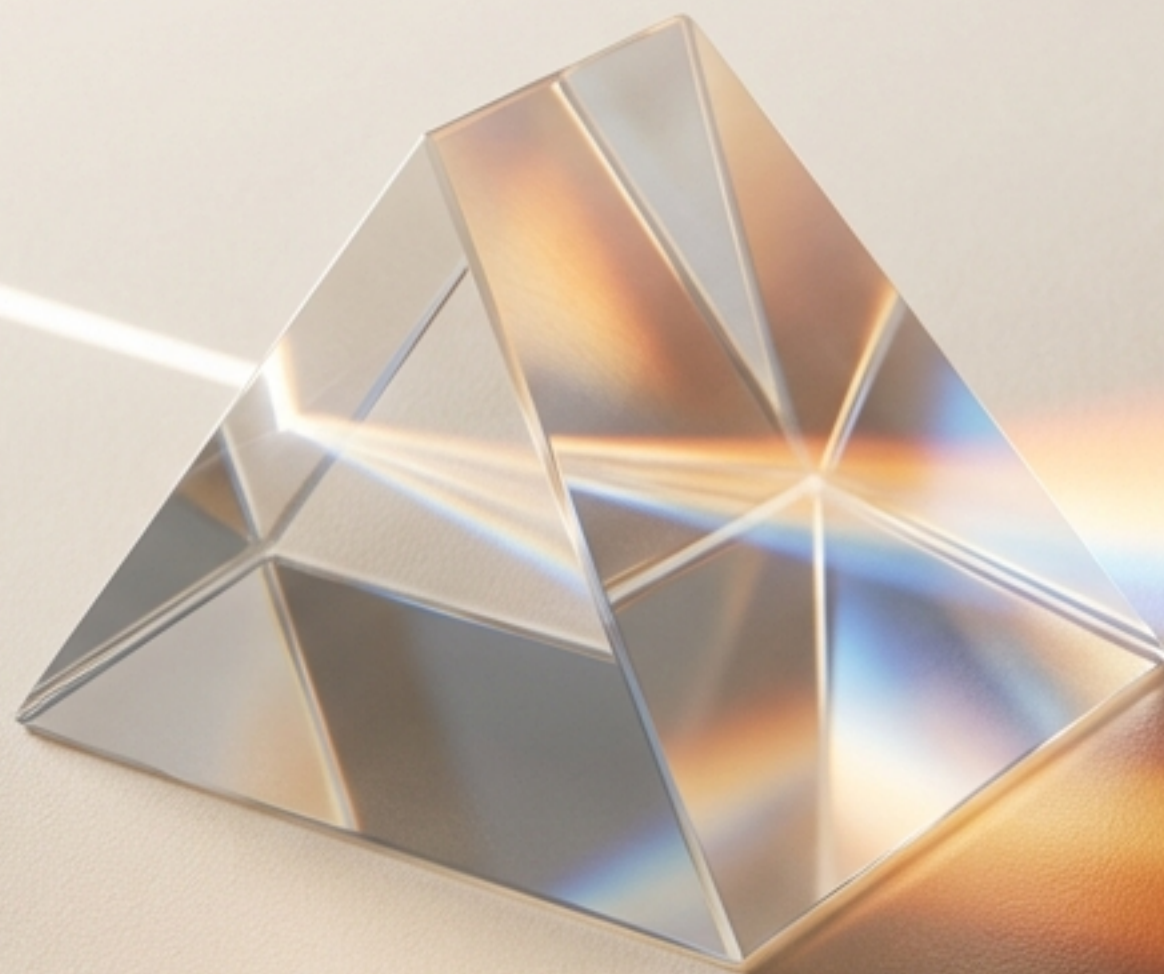
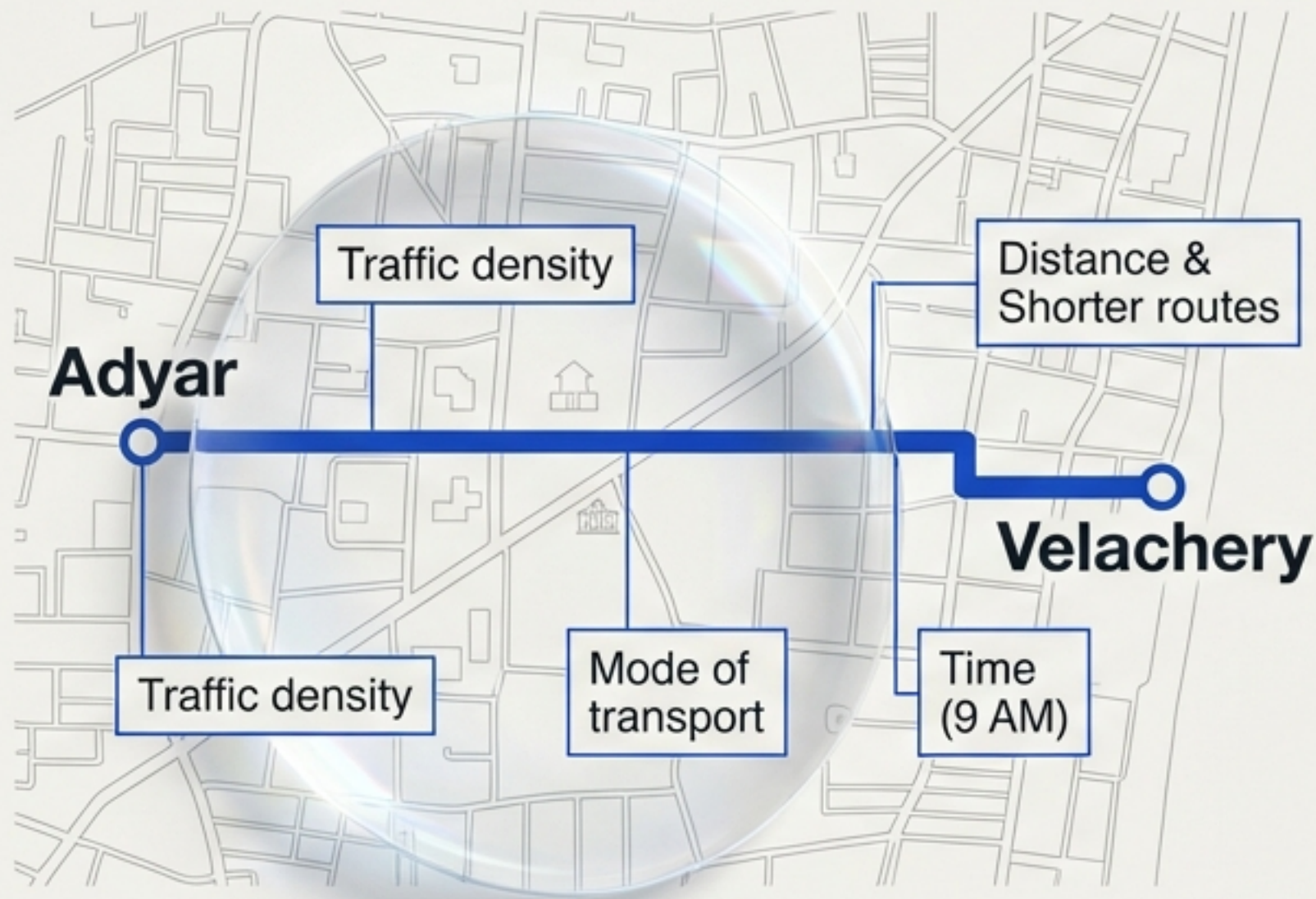


From Data to Generation

A Foundational Blueprint to Machine Learning,
Deep Learning, and Generative AI



Turning Millions of Records into a Single Truth



March 28, 1990: 29°C

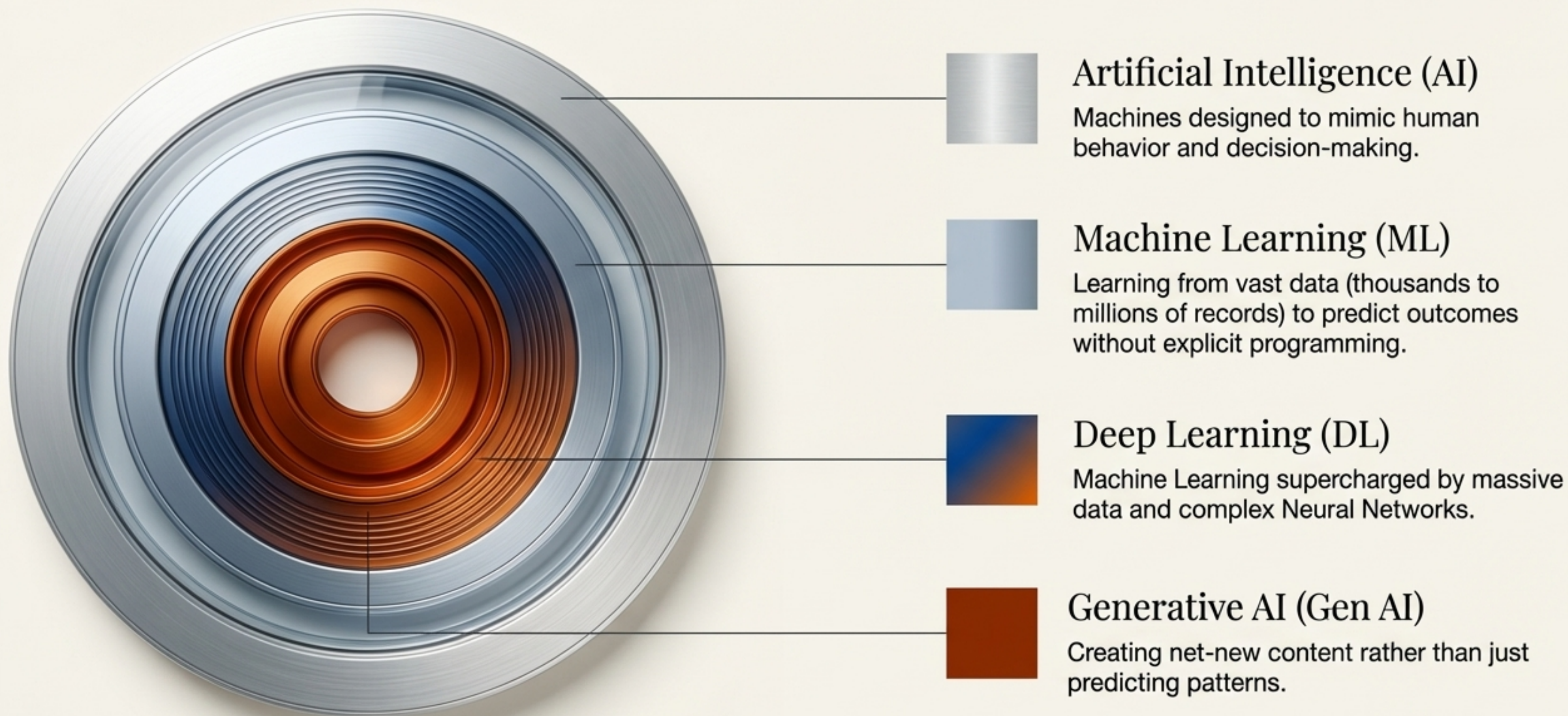
March 28, 2024: 31°C

March 28, 2025: 30°C

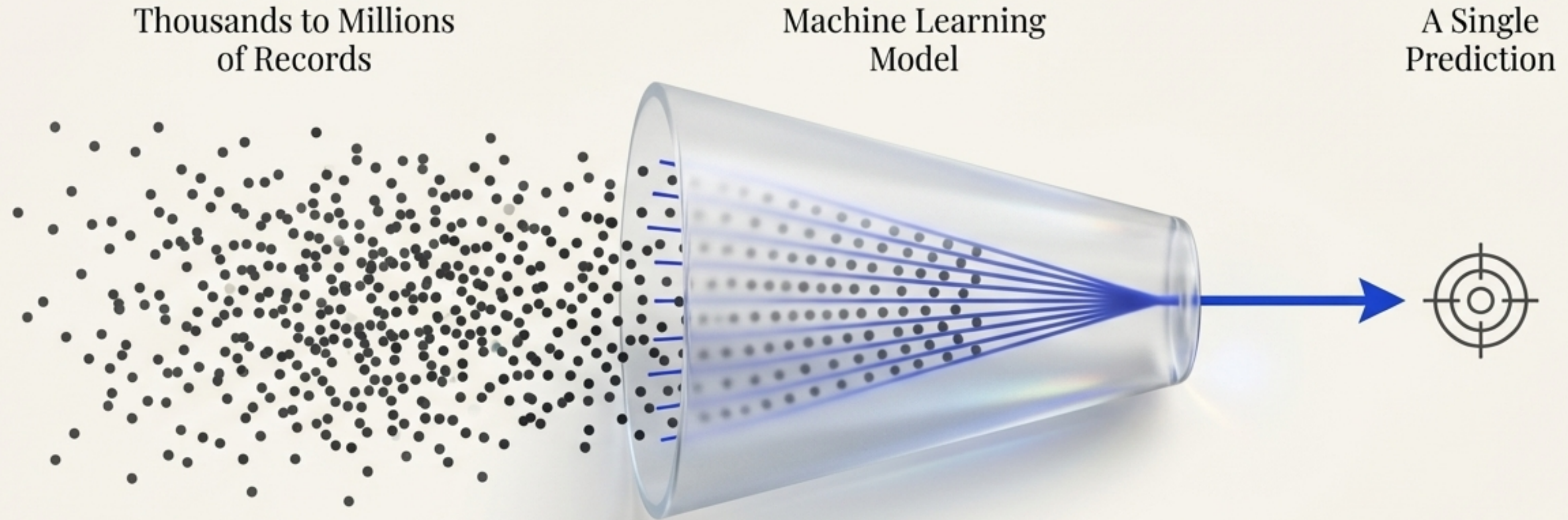
30°C

Current Situation + Millions of Previous Records = One Prediction.
Machines mimic human deduction by processing historical data faster than humanly possible.

The Nested Layers of Artificial Intelligence



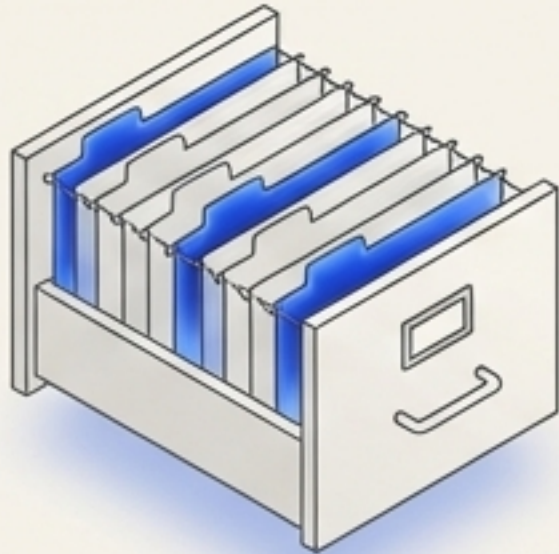
Machine Learning Focuses Chaos into Prediction



Machine Learning is a subset of AI that digests vast amounts of historical data. Its primary, defining output is a Prediction.

How Machines Learn from Data

Supervised Learning



Labeled Data

Mechanism: The model knows the target answer it is looking for and is trained on explicit labels.

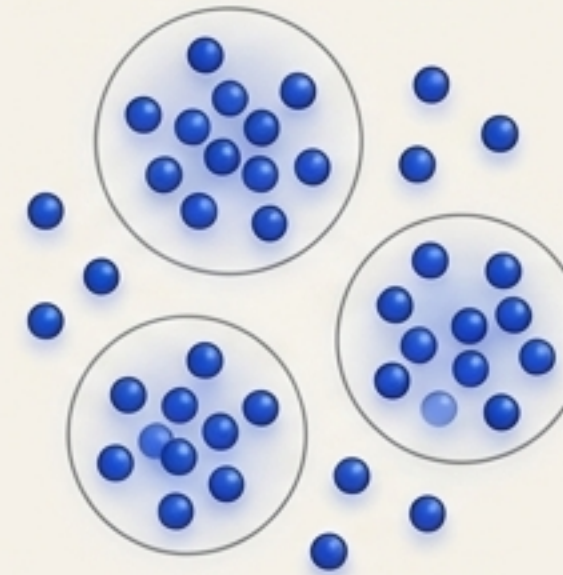
Type A: Linear Regression

Outputs a numerical value.
Example: Predicting Real Estate pricing based on home features.

Type B: Classification

Outputs categories or labels.
Example: Predicting the risk of a heart attack (Yes or No).

Unsupervised Learning



Unlabeled Data

Mechanism: The model does not know the target. It learns by finding hidden patterns, structures, and behaviors.

Example: Collaborative Filtering

Grouping users by genre preferences based on underlying behavior, even without explicit 'likes' or 'dislikes'.

Predicting Outcomes Across Different Data Types

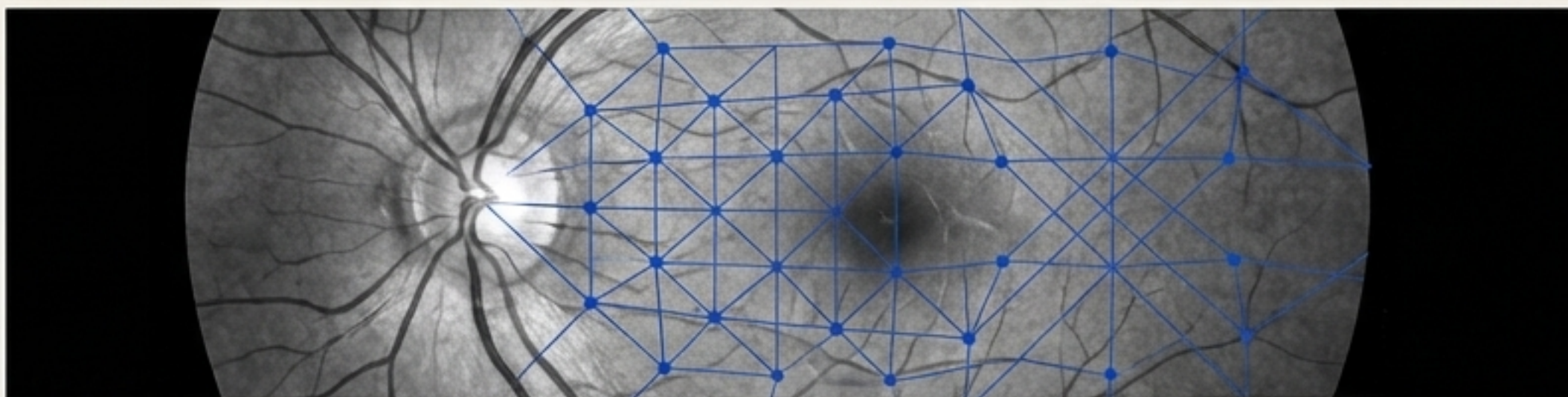
Structured Data

Age	Blood Group	Diabetic	Blood Pressure	Genetics	Smoker	Drinker
45	A+	No	120/80	Clean	No	Moderate
62	O-	Yes	140/90	History	Yes	Daily
38	B+	No	115/75	Clean	No	Rare



Heart Disease Risk:
YES / NO

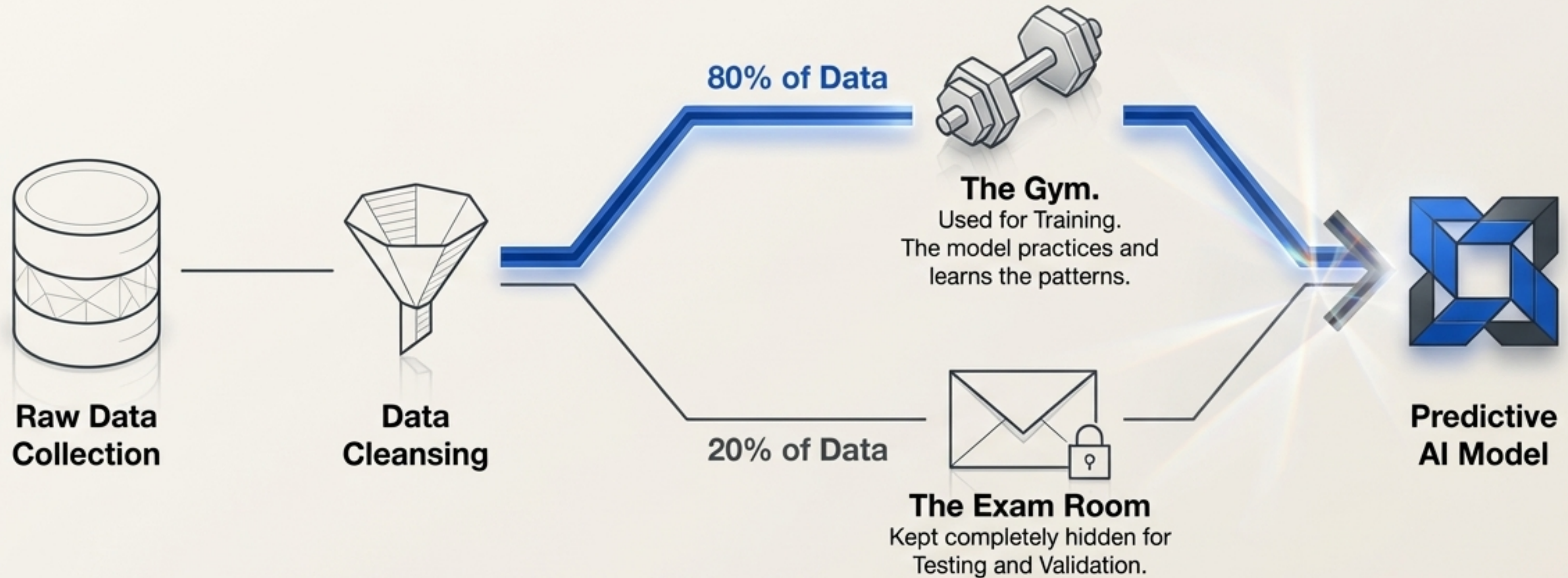
Unstructured Data



Risk Assessment: **HIGH**

Whether digesting millions of tabular rows or processing complex images through a CNN, the machine learning model evaluates brand-new, unseen data to predict risk accurately.

The Assembly Line of Predictive Models



Input for testing must be unseen data. If a model sees the exam questions while studying, we cannot trust its predictions.

Scaling Up with Deep Learning and Neural Networks

The Definition

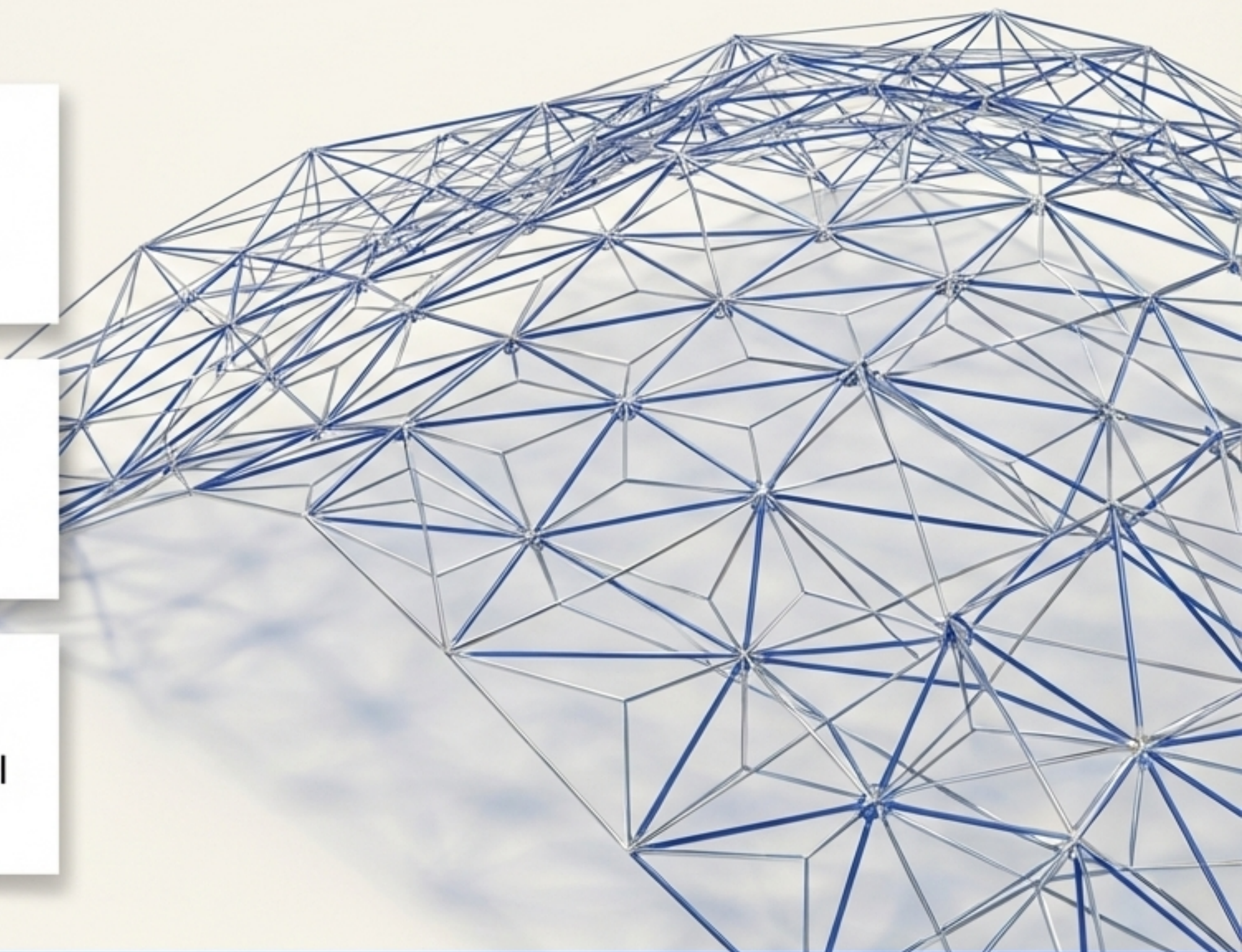
Deep Learning = Machine Learning at massive scale + Neural Networks.

The Scale

Moving from thousands of records to millions and trillions of interconnected data points.

The Capability

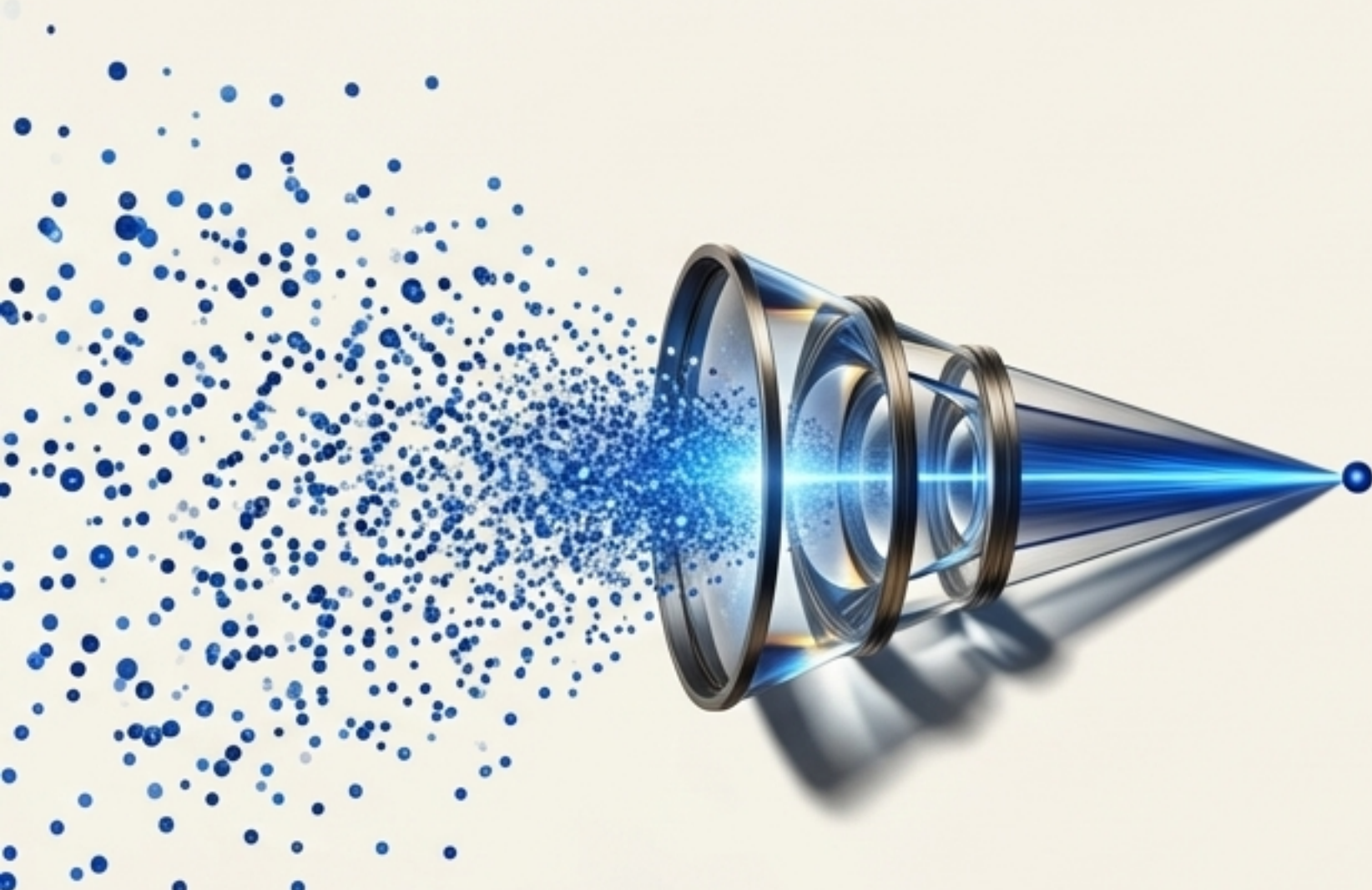
Deep Learning handles highly complex, multi-modal inputs (Image, Text, Video) simultaneously.



Deep Learning remains highly advanced, but its core function is still Predictive.

The Paradigm Shift from Analysis to Creation

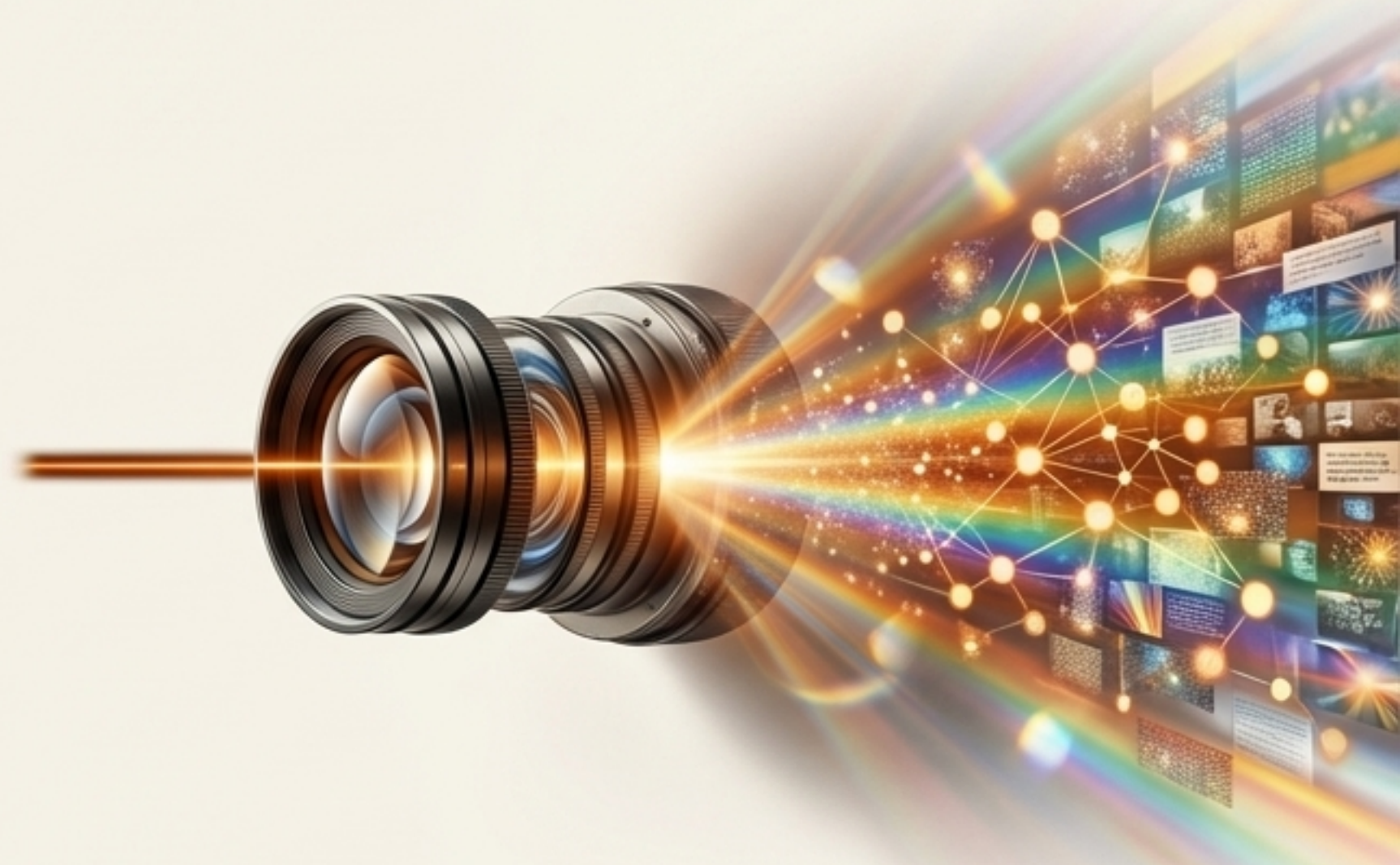
Predictive AI (ML/DL)



The Prism in Reverse

Analyzes existing historical records to forecast a single outcome.

Generative AI



The Projector

Uses deep neural networks to synthesize entirely net-new content.

Code

```
Terminal
import <odis.h>

class NeatFFvraoe {
  software.neetnent()

  def restoretenesefaresané, lones) {
    active > 11
    vñelocrvetizess5im) {
      concete.fagt?e1f1;
      printtato:segtl"Vode; not comea:"");
    }
  }
  return 0
}
```

Text

Headline Report About Reemil et Noose

...the ...

- The ...
- The ...
- The ...

	Score 1	Score 2	Score 3
Item 1	1.00	0.00	0.00
Item 2	0.00	1.00	0.00
Item 3	0.00	0.00	1.00
Item 4	0.00	0.00	0.00

Generative AI (Subset of DL)

Image



Video & Audio

Generative AI is completely untethered from simple predictions. It is an engine of creation.

The Anatomy of Large Language Models

G

Generative

Creates new content.

P

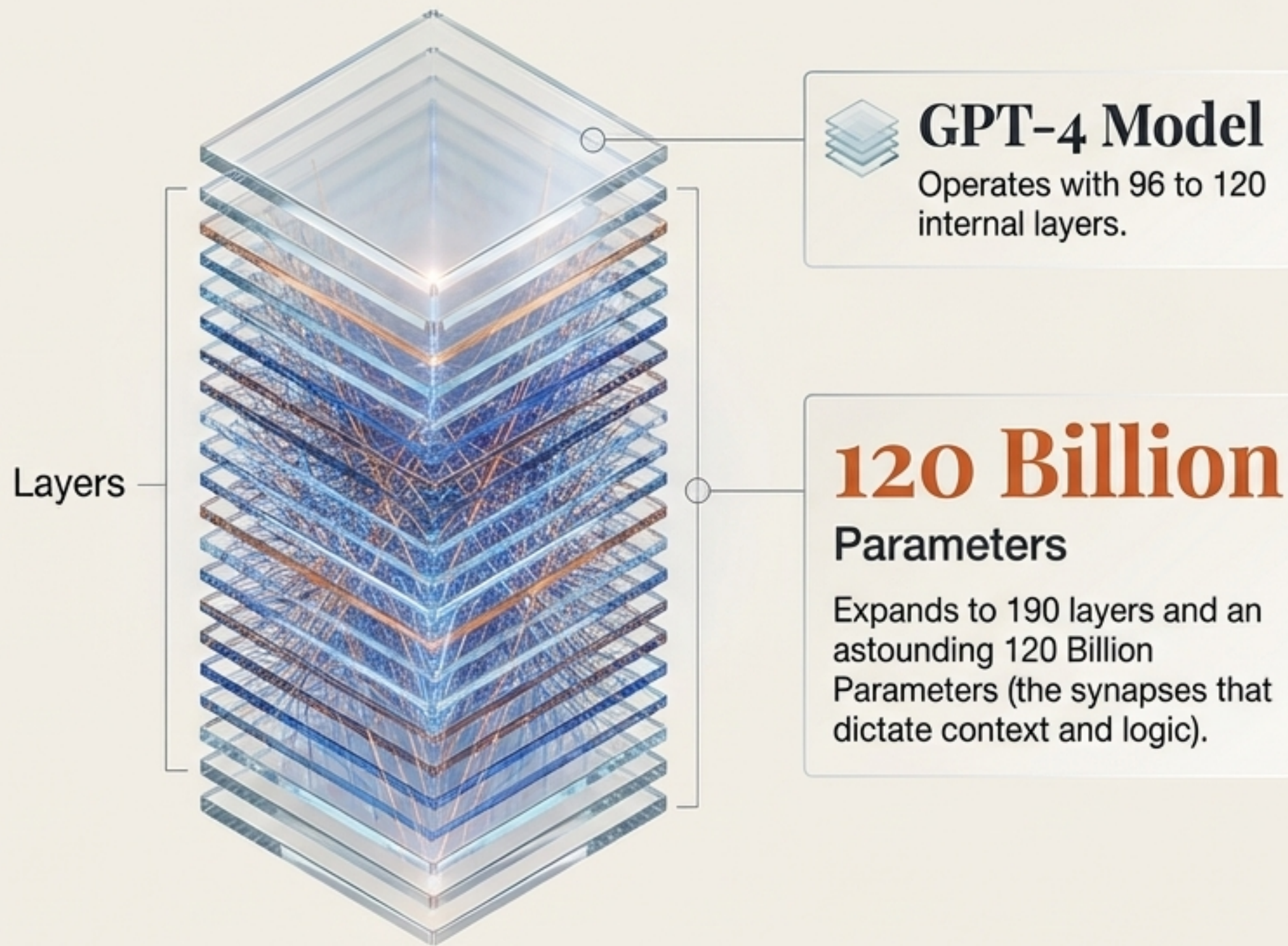
Pre-trained

Fed massive, universal datasets spanning software to humanities.

T

Transformer

The advanced neural architecture.



OpenAI's ChatGPT operates on models of unprecedented size.

Distilling the AI Landscape

	Machine Learning	Deep Learning	Generative AI
Core Mechanism	Predictive (Finding patterns)	Predictive (Finding complex patterns via Neural Networks)	Creative (Synthesizing new data)
Data Scale	Thousands to millions of rows	Millions to trillions of records	Massive universal pre-trained datasets
Primary Output	Numbers (Regression) or Categories (Classification)	Multi-modal analysis and risk assessments	Net-new Code, Text, Images, Video
Real-World Example	Predicting Real Estate Prices	Analyzing Retinal Scans for Heart Disease	ChatGPT drafting software architecture